



Lab 7. Facial Recognition

Overview

Develop a real-time system that uses a webcam to detect and recognise people, display their names on the video, and record their entry and exit times in a CSV file.

Tools Used

This lab uses the following libraries:

- OpenCV: OpenCV is a widely used computer vision library for handling images and video streams. In this lab, it is used to capture video from the webcam and display frames in real time.
- DeepFace: DeepFace is a high-level face recognition library that uses deep learning models to compare and recognise faces based on similarity.

Install the required libraries using:

```
pip install opencv-python  
pip install deepface
```

Reference:

- OpenCV: https://docs.opencv.org/4.x/dd/d43/tutorial_py_video_display.html
- DeepFace: <https://github.com/serengil/deepface>

Requirements

1. Prepare a face database by creating a folder structure where each person has their own folder containing one or more images of their face. The folder name represents the person's identity and will be used as the label during recognition.

```
faces_db/  
├── Amy/  
│   ├── front_view.jpg  
│   ├── side_profile.jpg  
│   └── smiling.jpg  
├── Bill/  
│   ├── front_view.jpg  
│   └── with_glasses.jpg
```

2. Use a webcam to capture a continuous video stream. The system should detect and recognise people in the video. For each recognised person, their name should be displayed on the video.
3. The system should identify when a person appears in the camera view and when a person leaves. It should maintain a log file in CSV format that records all entry and exit events with date and time. Each record must include the person's name, the event type (ENTER or EXIT), and the corresponding timestamp.

Name	Event	Time
Amy	ENTER	24/03/2026 16:33
Bill	ENTER	24/03/2026 16:35
Amy	EXIT	24/03/2026 16:36

Submission

Submit your source code via Canvas. You are also required to demonstrate your work to the tutor during the lab session.